

Mersenne TRUE GRAMMAR — Complete

Author: Martin Doina | $p_e = m^2 \times (p_a \times p_b) \pm k \mid 4$ k-Types

$p_e = m^2 \times (p_a \times p_b) \pm k$

diff: $p_c - p_d$ 2p+p: $2p_c + p_d$ 2p-p: $2p_c - p_d$ single: p_c

49/50

Found

1

Missing

11

diff

24

2p+p

14

2p-p

0

single

Done: 49/50

M#	p_e	m	m^2	p_a	p_b	$p_a \times p_b$	$m^2 \times \text{Base}$	k	k-Type	p_c	p_d	Formula	✓
M3	5	1	1	2	2	4	4	1	2p-p	2	3	$1^2 \times (2 \times 2) + 1$	✓
M4	7	1	1	2	2	4	4	3	2p-p	3	3	$1^2 \times (2 \times 2) + 3$	✓
M5	13	1	1	2	2	4	4	9	2p+p	2	5	$1^2 \times (2 \times 2) + 9$	✓
M6	17	1	1	2	2	4	4	13	2p+p	3	7	$1^2 \times (2 \times 2) + 13$	✓
M7	19	1	1	2	2	4	4	15	2p+p	5	5	$1^2 \times (2 \times 2) + 15$	✓
M8	31	1	1	2	2	4	4	27	2p+p	5	17	$1^2 \times (2 \times 2) + 27$	✓
M9	61	1	1	2	2	4	4	57	2p+p	13	31	$1^2 \times (2 \times 2) + 57$	✓
M10	89	1	1	2	5	10	10	79	2p+p	31	17	$1^2 \times (2 \times 5) + 79$	✓
M11	107	1	1	2	2	4	4	103	2p+p	7	89	$1^2 \times (2 \times 2) + 103$	✓
M12	127	1	1	2	2	4	4	123	2p+p	17	89	$1^2 \times (2 \times 2) + 123$	✓
M13	521	1	1	2	89	178	178	343	2p+p	127	89	$1^2 \times (2 \times 89) + 343$	✓
M14	607	1	1	3	31	93	93	514	diff	521	7	$1^2 \times (3 \times 31) + 514$	✓
M15	1,279	1	1	2	2	4	4	1,275	2p+p	607	61	$1^2 \times (2 \times 2) + 1275$	✓
M16	2,203	1	1	17	127	2,159	2,159	44	diff	61	17	$1^2 \times (17 \times 127) + 44$	✓
M17	2,281	1	1	3	607	1,821	1,821	460	diff	521	61	$1^2 \times (3 \times 607) + 460$	✓
M18	3,217	1	1	31	107	3,317	3,317	-100	diff	7	107	$1^2 \times (31 \times 107) + -100$	✓
M19	4,253	1	1	5	607	3,035	3,035	1,218	diff	1,279	61	$1^2 \times (5 \times 607) + 1218$	✓
M20	4,423	1	1	2	2	4	4	4,419	2p+p	2,203	13	$1^2 \times (2 \times 2) + 4419$	✓
M21	9,689	1	1	2	19	38	38	9,651	2p+p	3,217	3,217	$1^2 \times (2 \times 19) + 9651$	✓
M22	9,941	1	1	2	19	38	38	9,903	2p+p	107	9,689	$1^2 \times (2 \times 19) + 9903$	✓
M23	11,213	1	1	13	127	1,651	1,651	9,562	diff	9,689	127	$1^2 \times (13 \times 127) + 9562$	✓
M24	19,937	1	1	2	19	38	38	19,899	2p+p	9,689	521	$1^2 \times (2 \times 19) + 19899$	✓
M25	21,701	1	1	3	607	1,821	1,821	19,880	2p-p	9,941	2	$1^2 \times (3 \times 607) + 19880$	✓
M26	23,209	1	1	17	89	1,513	1,513	21,696	diff	21,701	5	$1^2 \times (17 \times 89) + 21696$	✓
M27	44,497	1	1	2	2,281	4,562	4,562	39,935	2p+p	19,937	61	$1^2 \times (2 \times 2281) + 39935$	✓
M28	86,243	1	1	19	4,253	80,807	80,807	5,436	diff	9,689	4,253	$1^2 \times (19 \times 4253) + 5436$	✓
M29	110,503	133	17,689	2	2	4	70,756	39,747	2p-p	19,937	127	$133^2 \times (2 \times 2) + 39747$	✓
M30	132,049	1	1	107	521	55,747	55,747	76,302	diff	86,243	9,941	$1^2 \times (107 \times 521) + 76302$	✓
M31	216,091	223	49,729	2	2	4	198,916	17,175	2p-p	9,689	2,203	$223^2 \times (2 \times 2) + 17175$	✓
M32	756,839	409	167,281	2	2	4	669,124	87,715	2p-p	44,497	1,279	$409^2 \times (2 \times 2) + 87715$	✓
M33	859,433	152	23,104	2	13	26	600,704	258,729	2p+p	86,243	86,243	$152^2 \times (2 \times 13) + 258729$	✓
M34	1,257,787	551	303,601	2	2	4	1,214,404	43,383	2p-p	21,701	19	$551^2 \times (2 \times 2) + 43383$	✓
M35	1,398,269	316	99,856	2	7	14	1,397,984	285	2p+p	89	107	$316^2 \times (2 \times 7) + 285$	✓

M#	p _e	m	m ²	p _a	p _b	p _a ×p _b	m ² ×Base	k	k-Type	p _c	p _d	Formula	✓
M36	2,976,221	853	727,609	2	2	4	2,910,436	65,785	2p-p	44,497	23,209	853 ² ×(2×2) + 65785	✓
M37	3,021,377	347	120,409	5	5	25	3,010,225	11,152	diff	11,213	61	347 ² ×(5×5) + 11152	✓
M38	6,972,593	1078	1,162,084	2	3	6	6,972,504	89	2p-p	89	89	1078 ² ×(2×3) + 89	✓
M39	13,466,917	466	217,156	2	31	62	13,463,672	3,245	2p+p	521	2,203	466 ² ×(2×31) + 3245	✓
M40	20,996,011	1449	2,099,601	2	5	10	20,996,010	1	2p-p	2	3	1449 ² ×(2×5) + 1	✓
M41	24,036,583	233	54,289	7	61	427	23,181,403	855,180	diff	859,433	4,253	233 ² ×(7×61) + 855180	✓
M42	25,964,951	✗ NOT FOUND (m 5000)											
M43	30,402,457	2753	7,579,009	2	2	4	30,316,036	86,421	2p+p	89	86,243	2753 ² ×(2×2) + 86421	✓
M44	32,582,657	41	1,681	2	9,689	19,378	32,574,418	8,239	2p-p	4,423	607	41 ² ×(2×9689) + 8239	✓
M45	37,156,667	3051	9,308,601	2	2	4	37,234,404	-77,737	2p-p	4,253	86,243	3051 ² ×(2×2) + -77737	✓
M46	42,643,801	3236	10,471,696	2	2	4	41,886,784	757,017	2p+p	89	756,839	3236 ² ×(2×2) + 757017	✓
M47	43,112,609	2075	4,305,625	2	5	10	43,056,250	56,359	2p+p	23,209	9,941	2075 ² ×(2×5) + 56359	✓
M48	57,885,161	2033	4,133,089	2	7	14	57,863,246	21,915	2p+p	107	21,701	2033 ² ×(2×7) + 21915	✓
M49	74,207,281	1141	1,301,881	3	19	57	74,207,217	64	2p+p	31	2	1141 ² ×(3×19) + 64	✓
M50	77,232,917	2779	7,722,841	2	5	10	77,228,410	4,507	2p+p	127	4,253	2779 ² ×(2×5) + 4507	✓
M51	82,589,933	76	5,776	3	4,423	13,269	76,641,744	5,948,189	2p-p	2,976,221	4,253	76 ² ×(3×4423) + 5948189	✓
M52	136,279,841	1899	3,606,201	2	19	38	137,035,638	-755,797	2p-p	521	756,839	1899 ² ×(2×19) + -755797	✓

49/50 FOUND Missing: M42